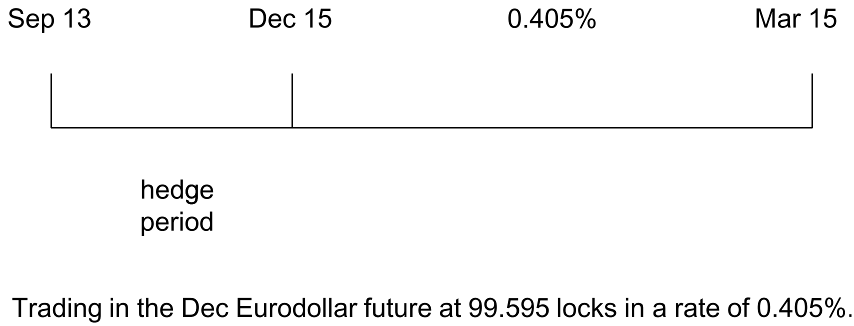

Figure 3.1: Hedging with a Eurodollar Future

By trading in Eurodollar futures, an investor is able to lock in the rate implied by the contract. For example, suppose that on September 13, 2010, an investor buys 100 of the Dec 10 Eurodollar contract (whose last trading day is Monday, December 13, 2010) at 99.595. If the investor holds the contract to expiration, he will lock in an interest rate for the 90-day period between December 15, 2010 and March 15, 2011 of $100 - 99.595 = 0.405\%$ and protect himself from a decline in the then-prevailing 3-month LIBOR rate below the level of 0.405% (see Figure 3.1). On December 13, the investor may enter into a Eurodollar investment at the then-prevailing 3-month LIBOR rate. Any profits or losses in the futures contract will exactly offset any changes in 3-month LIBOR from what the market is currently predicting. For example, suppose that on December 13, 3-month LIBOR sets at 0.305%. In this case the investor will earn interest of

$$\$100\text{mm} \times 0.305\% \times 90/360 = \$76{,}250. \tag{3.1}$$

In addition, the investor will make money on his position in futures. Since LIBOR settles at 0.305% at the settlement of the contract, the contract closes at 99.695. Thus, the position in futures earns

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rates. To see this, assume futures prices are positively correlated with interest rates and consider someone who is long a futures contract. When interest rates go up, futures prices will tend to go up, and the long will receive cash flow in the form of daily variation margin. When interest rates go down, futures prices will tend to go down, and the long will have to pay out cash. Thus, the long will be able to reinvest cash flow coming in from variation margin when rates are high and borrow to fulfill variation margin at low rates. This is an advantageous position for the long to be in, certainly as compared to the long in a comparable forward contract. Thus, in order to make a long indifferent between the two contracts, the futures price must be greater than the forward price. Sundaresan (1991) has looked at these issues extensively for the case of forwards and futures on interest rates. In summary, the assumption that we are making in this section (i.e., futures are the same as forwards) is not entirely innocuous. Nevertheless, it is suitable for our purposes here.

$$100 \times \$25 \times (9969.5 - 9959.5) = \$25{,}000. \tag{3.2}$$

Thus, the total earned is \$76,250 + \$25,000 = \$101,250. Notice that if the investor had been able to invest at a rate of 0.405% instead of the then-prevailing rate of 0.305%, he would have earned interest of

$$\$100\text{mm} \times 0.405\% \times 90/360 = \$101{,}250, \tag{3.3}$$

the same amount obtained by investing at the lower rate of 0.305% and benefiting from the position in futures (i.e., the sum of the quantities in Equations 3.1 and 3.2).

Conversely, if rates had gone up to, say, 0.505%, the investor would have again locked in a rate of 0.405% if he had hedged with futures. In this case the investor, upon investing his funds in December at a rate of 0.505%, will earn interest of

$$\$100\text{mm} \times 0.505\% \times 90/360 = \$126{,}250. \tag{3.4}$$

But this time the investor will lose money on his position in futures. Since LIBOR settles at 0.505% at the settlement of the contract, the contract closes at 99.495. Thus, the position in futures earns

$$100 \times \$25 \times (9949.5 - 9959.5) = -\$25{,}000, \tag{3.5}$$

a loss for the investor. Thus, the total earned is \$126,250 − \$25,000 = \$101,250, the sum of the quantities in Equation 3.4 and 3.5. This is the same amount shown in Equation 3.3, as if the investor had been able to invest at a rate of 0.405% instead of the then-prevailing rate of 0.505%. Thus, in the case of either rates going up or rates going down, the investor locks in a rate of 0.405%, the rate implied by the Eurodollar contract.[5]

3.1.1 The Link Between Swap Rates and Eurodollar Futures

Following Sundaresan (2009), let's consider another (somewhat contrived) example now. Suppose it is Monday, September 13, 2010, and a company

[5] We should be a little careful here. Although the nominal cash flows obtained by hedging and investing at the then-prevailing rate are the same as if we had invested at the rate implied by the Eurodollar futures, the timing of the cash flows differs. In our example, any investment in a Eurodollar time deposit will generate cash flows in March. But any position in futures earns daily variation margin from when we first enter the contract in September until the settlement of the contract in December. Even if we pretended the futures were forwards and paid out only at settlement, they would in that case pay in December, whereas interest in the Eurodollar deposit will pay in March. We will ignore such timing differences for our purposes here.

	Futures Prices	Implied Forward Rate	Current 3-month LIBOR
Sep 10	n.a.		0.290%
Dec 10	99.595	0.405%	
Mar 11	99.530	0.470%	
Jun 11	99.470	0.530%	

Table 3.1: Spot LIBOR and Futures Prices as of September 13, 2010

has a floating rate liability over the next year that resets quarterly indexed to 3-month LIBOR on a face amount of $100mm. The liability commences today based on today's 3-month LIBOR setting, which is 0.29%, and the remaining three reset dates correspond to the settlement dates of the Dec 10, Mar 11, and Jun 11 futures contracts. We will assume, for simplicity, that the liability for each quarter will accrue for 90 days (even though each quarter may not have exactly 90 days). The company is expecting an increase in LIBOR over the coming year and wants to lock in its liabilities from floating to fixed using Eurodollar futures. Table 3.1 provides information regarding futures prices for the three contracts as of September 13, 2010.

Can the company do anything to hedge its first three-month liability based off of today's 3-month LIBOR setting?[6] What can the company do to hedge its remaining exposure for the other three quarters? The company can sell a *strip* of 100 Eurodollar futures contracts to hedge against a rise in rates. By selling 100 of each of the Dec 10, Mar 11, and Jun 11 Eurodollar futures, the company will be locking in the rate implied in the respective futures contracts and will thereby protect itself against a rise in rates during each of the three subsequent quarters in which it is exposed. Assuming the company does hedge by selling the strip of Eurodollar futures, the cash flows it will lock in at the end of each quarter are shown in Table 3.2.[7]

That is, the final column in Table 3.2 shows the net cash flow that occurs from the combination of the liability and the futures position. More important, it shows the effective rate that the company has paid each quarter. For example, for the three-month payment that is due in June 2011, the company will pay a net amount of $117,500, and it is as if the company has paid a rate of 0.47% for the quarter. That is, the company was able to lock in the rate implied by the Mar 11 futures contract for the quarter.

[6]The answer is no. The rate has already set, and the cash flow that the company will owe for this quarter is already determined.

[7]Once again, there is actually a time mismatch between when the futures settle and when the company pays the cash on its quarterly liability, but we will ignore that here.

Futures Used to Hedge	Liability Payment Date	Locked in Rate	Cash Flow
	t_1 = Dec 13, 2010	0.290%	\$100mm × 0.290%/4 = \$72,500
Dec 10 @ 99.595	t_2 = Mar 14, 2011	0.405%	\$100mm × 0.405%/4 = \$101,250
Mar 11 @ 99.530	t_3 = Jun 13, 2011	0.470%	\$100mm × 0.470%/4 = \$117,500
Jun 11 @ 99.470	t_4 = Sep 19, 2011	0.530%	\$100mm × 0.530%/4 = \$132,500

Table 3.2: Cash Flows After Hedging with Eurodollar Futures

So let's take a step back now. The company had a floating rate liability and, without spending any cash today, was able to transform its liability from a floating one to a fixed one. The fixed rate of course varies from period to period (in accordance with the rates implied by the futures contracts), but nevertheless it is a fixed rate.

Can we find a single fixed rate such that when that rate is paid on the same dates as the floating payments, it has the same value as the rates implied by the futures contracts? Does such a rate exist? Sure it does. Now let's go about finding it. To start, let's compute the *discount factors* corresponding to each of the quarterly payment dates. The *discount factor* for time t, denoted $D(0,t)$, is that number that is multiplied by cash flow received at time t to compute its present value.[8] They are implied by today's LIBOR and the futures contracts. We will compute discount factors for times $t_1, \ldots, t_4$, as defined in Table 3.2. The discount factors are computed as follows: the first discount factor for cash flow received on December 13, 2010, is given by

$$D(0,t_1) = \frac{1}{1 + \frac{89}{360} \times 0.29\%} = 0.999284,$$

since there are 89 days between September 15, 2010, and December 13, 2010. The second discount factor is given by

$$D(0,t_2) = \frac{1}{1 + \frac{91}{360} \times 0.405\%} \times D(0,t_1) = 0.998262,$$

since there are 91 days between December 13, 2010 and March 14, 2011. The remaining two discount factors are computed in a similar fashion, as shown in Table 3.3.

[8]We will adopt the convention that discount rates are calculated so that the present value of cash flow is calculated as of the spot date of September 15, 2010, not the trade date of September 13, 2010.

Payment Dates	Actual Days	Implied LIBOR	Discount Factors
Mon, Dec 13, 2010	89	0.290%	0.999284
Mon, Mar 14, 2011	91	0.405%	0.998262
Mon, Jun 13, 2011	91	0.470%	0.997077
Mon, Sep 19, 2011	98	0.530%	0.995641

Table 3.3: Discount Factors Implied by Eurodollar Futures and Today's Spot LIBOR Setting Assuming a Spot Date of Wednesday, September 15, 2010

Payment Date	Implied Rate	Discount Factor	Locked In Flow	PV Locked in	Fixed	PV Fixed
13-Dec-10	0.290%	0.999284	72,500	72,448	105,908	105,832
14-Mar-11	0.405%	0.998262	101,250	101,074	105,908	105,724
13-Jun-11	0.470%	0.997077	117,500	117,157	105,908	105,598
19-Sep-11	0.530%	0.995641	132,500	131,922	105,908	105,446
				422,601		422,601
		Fixed rate	0.424%			

Figure 3.2: Solving for the Fixed Rate

Now remember, we are looking for a single fixed rate that when paid on the same dates as the floating payments has the same present value as the four different fixed rates the company was able to lock in using the futures contracts. That's easy to find when we use a spreadsheet like the one in Figure 3.2. In this figure, we have iterated for a single fixed coupon (which turned out to be 0.424%) such that the discounted cash flows associated with this single fixed coupon have the same present value as the four distinct fixed coupons that the company could lock in by trading in the futures.[9]

[9] Here's some detail regarding some of the calculations in Figure 3.2. The implied rates and discount factors are taken from Table 3.3. The locked-in flows are the nominal cash flows we obtain by accruing the locked-in rate for 90 days on a notional of $100mm. These flows were actually determined previously in Table 3.2. We then seek to determine the single fixed rate such that we are indifferent between paying that fixed rate or paying

But then we're done. We found that the company was able to transform the nature of its liability from a floating one based upon LIBOR to a fixed one, in which the fixed rate that the company pays every quarter is the same. The company was able to do this by using futures and did not need to take any money out of its pocket today. The present value of the fixed flows based on a rate of 0.424% is equal to the present value of the floating flows based on LIBOR that the company is contracted to pay.

This gets us extremely close to the notion of computing the 1-year swap rate. After all, the 1-year swap rate is, by definition, the fixed rate in a 1-year swap such that swap is an NPV = 0 transaction to both sides. In other words, the 1-year swap rate is the fixed rate in a 1-year swap such that the present value of the fixed cash flows is equal to the present value of the floating cash flows. As we just showed in the example of our corporation, on September 13, 2010 the present value of the floating cash flows that the company is due to pay is equal to the present value of the fixed cash flows based on a rate of 0.424%. Is it fair to say that the 1-year swap rate must be this same 0.424%?

Not exactly. Suppose it is once again September 13, 2010, and consider now a spot starting 1-year swap in which both sides pay quarterly. The swap will be effective September 15, 2011, and in accordance with standard conventions will make quarterly payments on the dates shown in Table 3.4. But if we compare the payment dates in a standard 1-year swap to the dates on which the corporation made its payments, we see that they are not the same. The payment dates for the corporation were chosen to concur with the settlement date of the Eurodollar futures. The payment dates in a 1-year swap are slightly different. Moreover, the dates upon which we reset LIBOR in the swap differ from the reset dates of the corporation's floating liability, the latter of which were chosen to match up with the settlement dates of the Eurodollar futures contracts.

This means that the rate of 0.424% that we calculated will not exactly equal the 1-year swap rate. Again, the 1-year swap rate has LIBOR reset and payment dates that do not concur with the settlement dates of the Eurodollar futures contracts. This means that the calculation of the 1-year swap rate will not be as "clean" as was the example of the corporation transforming

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the individual locked-in rates that we obtained by using futures. To say we are indifferent means that the present value of both streams is the same. As indicated, we solve for a fixed rate of 0.424%. Given this fixed rate, the nominal fixed cash flows are equal to \$100mm $\times$ 0.424%/4 = \$105,908. The present value of each nominal cash flow is found by multiplying by the respective discount factor. For example, the present value of \$105,908 to be received on June 13, 2011 is equal to \$105,908 $\times$ 0.997077 = \$105,598. The fixed rate of 0.424% is such that the sum of the present value of the locked-in flows is equal to the sum of the present value of the fixed flows.

LIBOR Reset Date	Payment Dates
Monday, September 13, 2010	Wednesday, December 15, 2010
Monday, December 13, 2010	Tuesday, March 15, 2011
Friday, March 11, 2011	Wednesday, June 15, 2011
Monday, June 13, 2011	Thursday, September 15, 2011

Table 3.4: LIBOR Reset and Payment Dates in a Standard Spot Starting 1-Year Swap on September 13, 2010

the nature of its floating rate liability. For instance, what rate should we use to "transform" the floating rate in the swap that sets on March 11, 2011? Inasmuch as this date does not concur with the settlement date of the March 11 future, we will need to resort to using some sort of interpolated rate using implied rates from multiple futures. Similarly, we will need to adjust our discount factors since our payment dates do not concur with settlement dates of the Eurodollar futures. Using a proprietary swap model, the 1-year swap rate on September 13, 2010 (quoted quarterly bond versus 3-month LIBOR quarterly money priced using market data inputs as given in Table 3.1) is 0.443%.

The building of a swap model that incorporates these features requires much work and is a hot topic of discussion throughout the industry. In general, most swap models use a combination of deposit rates (much like we used the spot LIBOR setting of 0.29%), a series of Eurodollar futures and implied rates, and (due to the limitations of Eurodollars in terms of both liquidity and availability) swap rates that trade in the market. Much care is spent on interpolation techniques, and the creation of robust swap models is often considered as much an art as a science. Many general techniques used are well known throughout the industry, but many firms have developed proprietary techniques as well. A further discussion of these issues is beyond the scope of this discussion and our needs, but the interested reader is referred to Hagan (2003), Ron (2000), Zhou (2002), and Flavell (2010).

3.1.2 The Futures Convexity Bias

As we discussed in Section 3.1.1, Eurodollar futures are often used to price fixed-floating swaps. Swap dealers will often use these futures to hedge the interest rate exposure implicit in a swap, selling futures when they received fixed in a swap or buying futures when they pay fixed in a swap.[10] It turns

[10] In Section 2.2 we discussed that a dealer will sometimes use Treasuries to hedge a swap, and now we are indicating that a dealer may use Eurodollar futures to hedge. What's

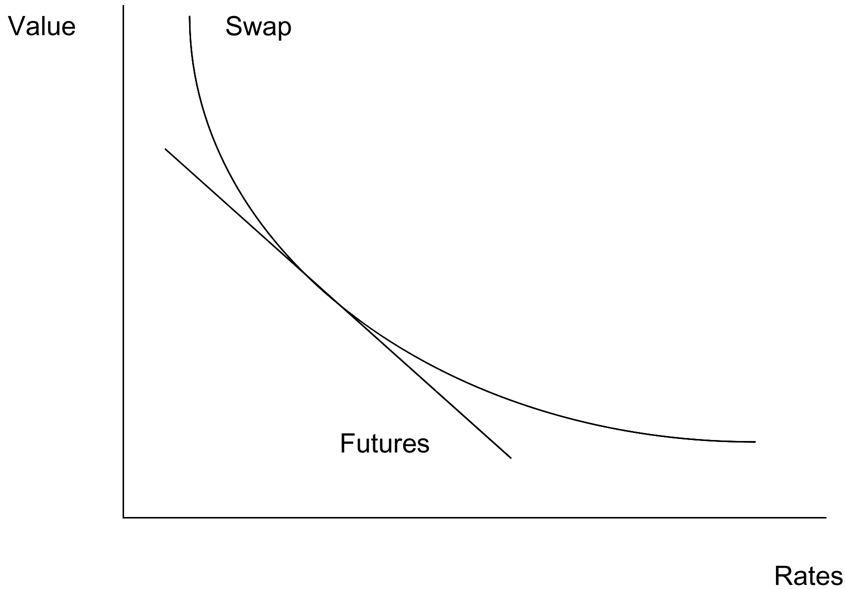

Figure 3.3: The Futures Convexity Bias

out, however, that the swap rates implied by pricing from Eurodollar futures are not equal to the swap rates observed in the market (even after adjusting for reset and payment mismatches between swaps and futures). The difference in these two rates is due to the *futures convexity bias.*

In order to gain insight into the nature of this bias, consider the following: in Section 2.1, we mentioned that receiving fixed in a swap is a positively convex position, since for a shift in swap rates up or down of equal magnitude the fixed rate receiver benefits more in a rally than he is hurt in a sell-off. But Eurodollar futures are nonconvex instruments: the change in value of a Eurodollar futures contract for a bp move in rates is always a constant \$25, regardless of the level of rates. These relationships are displayed in Figure 3.3. This convexity bias implies that there is a systematic benefit to receiving fixed in swaps and hedging by selling Eurodollar futures. If we receive fixed

the difference? A dealer who uses Treasuries to hedge a swap will be left with a spread position: a dealer who pays fixed and buys Treasuries will be long swap spreads, and one who receives fixed and sells Treasuries will be short spreads. On the other hand, a dealer who uses Eurodollar futures to hedge will not have a spread position. A dealer will make a choice of which instrument to use to hedge based on whether or not he wants a residual position in spreads as well as on the relative liquidity in Treasuries and Eurodollars at the time.

in swaps and can hedge our risk by selling futures, then we would benefit by being long a position that is positively convex and being short a position that is nonconvex. As Figure 3.3 demonstrates, the more volatile rates are, the greater the value of the convexity mismatch (i.e., the more the curve and the line diverge from one another).

The punch line here is that swap rates, as implied by Eurodollar futures, are too high. Since there is a systematic benefit from receiving fixed in a swap and selling futures, this means that the fixed rate receiver should be willing, everything else the same, to receive a lower fixed rate to compensate for the convexity benefit he receives (and which the fixed rate payer gives up). How much lower should the fixed rate be? In order to answer this question theoretically, we would need to employ a model of interest rate movements to put a value on this convexity. This is beyond the scope of our discussion; however, the interested reader is referred to Kirikos and Novak (1997) and Gupta and Subrahmanyam (2000). In general, the bias is an increasing function of the volatility of interest rates and of the maturity of the swap (longer maturity swaps exhibit greater convexity just like fixed rate bonds).

The conventional wisdom held by many in the market is that the convexity bias was ignored (actually not recognized, sotto voce) by swap traders in the 1980s and even early 1990s. An early analysis of the convexity bias in swaps was conducted by Burghardt and Hoskins (1995). They claim that although the swaps market had by the mid-1990s recognized the bias, swap rates trading in the market did not fully reflect the value of the convexity bias. In a later study, Gupta and Subrahmanyam (2000) indicate that in the period 1987–1990 the differential between swap rates in the market and rates implied by futures were insignificantly different from zero. They further indicate that during the period 1991–1993, the differential between the rates was significant (from a statistical point of view) and was very significant in the period 1994–1996. Their findings are in general agreement with the notion that the bias was ignored by traders in the early years of the swaps market and recognized in later years. They find, for example, that the value of the rate differential during the period 1991–1993 was averaging 8–10 bps for swaps of different maturities.

3.2 Moving On: Bootstrapping the Curve and Creating a Swap Model

In this section we are going to put the link between swaps and Eurodollar futures in our rearview mirror. From this point on, we are going to take it for granted that all of the swap rates (at least those on the short end of the curve) we see in the market on a composite screen like the one in Table 1.1 have a connection to Eurodollar futures prices.[11] We will from this point on regard swap rates as primitive objects. We will develop a technique known as *bootstrapping* in order to obtain zero coupon and forward rates implicit in the swap curve. Use of these rates will enable us to price a wide variety of trades.

Let's start by reminding ourselves of the following: we've previously alluded to the fact that we can think of receiving fixed in a swap as a long position in a fixed rate bond (whose value is denoted B_{fixed}) and a short position in a floating rate bond (whose value is denoted B_{floating}). The value of the swap is therefore

$$\text{Swap} = B_{\text{fixed}} - B_{\text{floating}}.$$

This interpretation is valid if we assume that the two counterparties exchange principal at maturity. Although the counterparties do not literally exchange principal at maturity, the value of the swap will not change if we assume that each counterparty both pays and receives an amount equal to the notional at maturity. The fictitious notional payment at maturity in conjunction with the fixed coupons constitutes a fixed rate bond. And the fictitious notional payment at maturity in conjunction with the floating rate coupons constitutes a floating rate bond.

Let's consider now this floating rate bond, which pays coupons equal to LIBOR. We want to compute the present value of these cash flows, using the swap curve for discounting purposes. Very loosely speaking, we are discounting these cash flows using the forward LIBOR curve. Hopefully this suggests that the present value of the bond is therefore equal to par, or the amount of the notional. We will show that this is true later on in this chapter, but we will accept it as fact for now.

This is a really important fact, so let's emphasize it one more time: *in a spot starting swap that is executed today, the present value of the floating side of the swap, including principal at maturity, is par. Moreover, on any of the dates on which LIBOR is reset, the value of the floating leg must also be worth par.*[12]

Since a swap is an NPV = 0 transaction when we first enter into it, and since the value of the floating side is worth par when we enter into it, this

[11] As mentioned previously, Eurodollar futures contracts don't trade past 10 years and thus don't explain swap rates beyond 10 years. Moreover, some traders indicate that they do not use Eurodollar futures beyond a few years to price swaps. Nevertheless, we will proceed taking the swap rates that trade in the market as a given. The forces of supply and demand ultimately determine where swap rates trade in the market.

[12] If we are not on a reset date, but rather in the middle of a period, the value of the floating side need not be par. Again, we will show this is the case later in this chapter.